Abdul Azeem Shaik

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Professional Summary

Software Engineer with 3+ years building scalable distributed systems and published researcher (Google AI funded, ACM JCSS). Currently researching adversarial attacks on multi-agentic AI systems in cybersecurity workflows at Georgia Tech's School of Cybersecurity and Privacy. Contributed to the mainnet launch of Shardeum, the world's first sharded autoscaling EVM compatible L1 blockchain (80M+ transactions, 150K+ validators), with open source contributions spanning Shardeum, Liberdus, Shardus, Cosmos, and Farcaster Snapchain; hands-on experience across distributed systems, AI/ML engineering, and cybersecurity, with coursework in Embedded Systems, Digital Electronics, and GPU Programming.

Education

Georgia Institute of Technology — USA
M.S. in Computer Science, Specialization: Artificial Intelligence (Expected) — *Dec 2026*

BITS Pilani — Hyderabad
B.E. in Electronics and Communication Engineering — *2023*

Experience

Georgia Institute of Technology — School of Cybersecurity and Privacy — USA
Researcher — AI Security — *Apr 2026 – Present*

- Investigating adversarial attacks on multi-agentic AI systems in cybersecurity workflows, targeting SOC pipelines with agentic orchestration and generative AI threat surfaces.
- Building evaluation pipelines to assess robustness of AI-driven SOC workflows against prompt injection, tool misuse, and inter-agent manipulation.

Liberdus — USA
Software Engineer — Distributed Systems (Remote) | TypeScript, Node.js, Rust, Python — *Jan 2026 – Present*

- **Cross-Chain Infrastructure:** Built the Liberdus BSC bridge and cross-chain payment platform, enabling asset transfers within a unified messaging and payments interface.
- **Performance:** Reduced transaction time to finality by 85% by resolving TSS signer deadlocks via entry synchronization fixes across signing parties; improved transaction latency by 38% through smarter RPC request routing across validator nodes.
- **Sync & Memory:** Cut historical data sync latency by 63% by introducing transaction gossip protocols into the sync pipeline; reduced heap memory usage by 50% and eliminated OOM crashes through RPC response caching.
- **Scalability:** Built a blockchain indexing microservice to make historical transaction sync horizontally scalable; resolved database contention in SQLite-backed components using betterSQLite concurrency best practices.
- **Cryptography & Consensus:** Implemented cryptographic libraries in Rust for identity verification and threshold signature support; built a quorum-based validator response mechanism in Rust for Byzantine-resilient consensus queries; added forward gateway functionality for TSS signers in Rust to enable bridge client communication.
- **DAO & Security:** Architected the distributed backend for Liberdus on-chain DAO governance system; audited bridge smart contracts across chains, identifying and remediating security vulnerabilities; led DevSecOps across Liberdus open-source repositories as the primary security maintainer.

Shardeum — USA
Senior Software Engineer (Remote) | Rust, TypeScript, Node.js — *Aug 2022 – Jan 2026*

- **Node Monitoring:** Built a horizontally scalable monitoring system handling 10,000+ req/sec across multiple cores and machines, with shared global state and real-time visibility across server components and dashboards.
- **Performance:** Reduced communication latency by 34% (82 ms → 54 ms) via process profiling in TCP library; boosted RPC throughput 60% with WebSocket data distribution; achieved 14× speedup on ECDSA verification (10 ms → 700 μs); reduced cryptographic latency 29% by migrating to `libsodium`.
- **Security & Consensus:** Took part in audits covering consensus safety, DDoS, MITM, and replay attacks; discovered and fixed critical bugs in Proof of Quorum consensus; improved failed-node detection for network stability under poor connectivity.
- **Platform:** Redesigned account storage to reduce memory footprint; built checkpoint-aware patching pipeline with fault-tolerant recovery; hardened transaction authorization end-to-end; built RPC reverse proxies and service test suites.
- **Infrastructure:** Built distributed executable caching across partitioned nodes; optimized gossip protocol with custom serialization; redesigned storage to eliminate redundant per-node memory; managed CI/CD with GitHub Actions and pm2.

BITS Pilani — Hyderabad
Researcher — HCI & Applied AI — *Aug 2021 – Jul 2022*

- **Publication:** "AI-based Market Intelligence Systems for Farmer Collectives", funded by Google AI for Social Good; published in the ACM Journal on Computing and Sustainable Societies (JCSS).

- Deployed a production Android app with integrated forecasting models; built Java/Kotlin backend with Firebase; validated with real market data and positive farmer feedback.

BITS Pilani Hyderabad
Researcher — HCI and Security *Dec 2021 – May 2022*

- Investigated security of low-cost Android devices via static/dynamic analysis and Wireshark PCAP traffic analysis; identified insecure cross-border communications, bloatware, and permission misuse.

BITS Pilani Hyderabad
Researcher — IoT and ML *Jan 2022 – May 2022*

- Surveyed ML optimization techniques for resource-constrained IoT environments under Prof. Nikumani Choudhary; focus on Federated Learning for decentralized training across edge devices without centralizing sensitive data.
- Configured IoT testbeds using Arduino Uno and Raspberry Pi to reproduce experiments from edge ML deployment literature; implemented distributed training scripts in Python across constrained hardware.

Open Source Contributions

Farcaster — Snapchain
Rust, RocksDB, Protocol Buffers, gRPC *github.com/farcasterxyz/snapchain*

- Enhanced event filtering and querying APIs with timestamp and block-based filters; designed efficient serialization to optimize network communication and database performance.

Liberdus — TSS Signer
Rust, TypeScript, Node.js *github.com/Liberdus/tss-signer*

- Built multi-network event processing pipeline with configurable routing, gas estimation, adaptive batch sizing, per-network rate limiting, and block-state persistence.
- Fixed state corruption where failed transactions overwrote completed ones; added response validation and async file writes for block-state persistence.
- Implemented Ed25519 signing/verification in Rust with `signature||message` format for library compatibility; added server-side signed request body validation.

References

Omar Syed
CTO, Shardus & Liberdus *linkedin.com/in/omarsyed1*

- Co-Founder of ARIMAA & Shardeum; Principal Architect at Zynga & Yahoo; Engineer at Raytheon & NASA

Marc Hanson
Co-Founder & Partner, Super Mega Lab *linkedin.com/in/marc-hanson-579469a*

- Former VP Engineering at Shardeum; CTO at BossFight; Director of Engineering at Zynga; Engineer at Microsoft

Technical Skills

Languages: Rust, TypeScript, Python, C++, Java
Core Expertise: Distributed systems, performance optimization, consensus protocols, cryptography
Backend & APIs: Node.js, Express.js, RESTful API, JWT, WebSockets, JSON-RPC, gRPC
Frontend: React.js, Vue.js, Nuxt.js, JavaScript
Databases: SQLite, PostgreSQL, Firebase
DevOps: Docker, Kubernetes, GitHub Actions, pm2, Protocol Buffers, Git
ML & Data: XGBoost, Pandas, NumPy, Matplotlib
Tools: Jest, Jupyter, VS Code, Claude Code, Cursor AI
Relevant Coursework: Machine Learning, NLP, Reinforcement Learning, Information Security, GPU Hardware and Software, Operating Systems, IoT, Microprocessors & Interfacing, Communication Systems & Networks, Control Systems